

Assignment on Functional Requirements

Final Functional Requirements

Proposed Project Title

Quivaultis: Mood-Based Suggestion Powered Game Store with Peer-to-Peer Trading and Smart Support

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Project Overview

Quivaultis is a web-based gaming platform that brings game discovery, comparison, purchasing, peer-to-peer trading, community feedback, personalized recommendation and customer support into one system. Customers can register and log in, browse the game catalogue, search and filter titles, open detailed game information, compare exactly two selected games, manage a cart and complete the current internal purchase flow. Games associated with a completed purchase are available from the customer's Owned Games Library.

The platform also supports peer-to-peer marketplace activity. Customers can create listings for games or skins, browse active listings, send offers, purchase eligible listings using wallet balance, manage inventory and review transaction history. The Community Review System is handled separately and allows customers to review games they own, provide star ratings, edit or delete their own reviews, mark reviews as helpful, react to community feedback and use search, filter and sorting tools. A dedicated Deals and Watchlist System shows currently discounted games, lets customers watch or unwatch deals and stores their price-drop alert preferences.

A major feature of Quivaultis is the Mood-Based Game Recommendation System. It evaluates the customer's selected mood, session time, preferred play style, difficulty and feedback such as Love, Played and Not for Me to rank suitable games. The system also includes customer profile management with profile completion and ranks, badge verification, support ticket handling, staff game management, wallet and marketplace transaction tracking, and admin user/platform management. Customer, Staff and Admin roles are separated through authenticated and role-based access.

Technology Stack

Component	Technology / Approach
Language	JavaScript
Frontend	React.js with Vite
Backend	Node.js and Express.js
Database / ODM	MongoDB with Mongoose
Authentication	JWT-based authentication
API Communication	REST API
Database Hosting	MongoDB Atlas
Styling	CSS with a modern dark gaming UI
Client-side State	LocalStorage for selected UI preferences, notifications and client-side state
Recommendation	Internal mood and preference matching logic

User Roles

Customer: Creates an account; browses, searches and compares games; manages cart and owned games; uses the marketplace and wallet; writes community reviews; uses the Deals watchlist; updates profile information; requests badges; receives mood-based suggestions; and submits support tickets.

Staff: Maintains game records, stock and discounts; handles support tickets; and reviews badge requests. Staff access is more privileged than customer access but is below admin access.

Admin: Performs centralized platform management including user and role management, game records, badge requests, support tickets, deals-related administration and operational overview information.

Functional Requirements - Common Workflows

SL	Common Workflow	Description
1	Registration and Login System	Users can create an account using username, email and password. Registered users can log in and receive a JWT-backed authenticated session.
2	Role-Based Access and Navigation	The system supports Customer, Staff and Admin roles. After login, users are directed to the appropriate interface and protected operations require an authenticated token; staff/admin-only actions are additionally restricted by role.

Implementation Scope and Assumptions

- The Store currently implements title search, multiple filters, game details, comparison of exactly two selected games, cart management and an internal purchase flow.
- A live external payment gateway is not part of the current implementation. SSLCommerz is retained as a feasible future integration for real payment confirmation.
- RAWG/IGDB and Cloudinary remain optional future integrations. The current system can operate using MongoDB game records and its existing image/file handling approach.
- Operational counts may appear in role dashboards, but a separate advanced analytics and sales-reporting module is not claimed as part of the current scope.

Security and Data Integrity Requirements

Area	Requirement
Authentication	Protected routes shall require a valid JWT token before returning private user data or allowing protected operations.
Authorization	Staff/Admin operations shall verify the user role on the backend and reject unauthorized requests.
Credential Protection	Environment secrets such as JWT secret and database credentials shall remain in environment configuration and shall not be exposed in frontend code or public repositories.
Password Hardening	For production deployment, stored passwords shall be protected using secure one-way password hashing rather than plain-text storage.
Input & Ownership Validation	The backend shall validate required fields and verify ownership/eligibility before sensitive actions such as review updates, marketplace operations, ticket access and badge requests.
Database Access	MongoDB Atlas network/database access shall be restricted to authorized users and appropriate network rules for deployment.

External API / Service Technical Feasibility

The following services were evaluated for technical feasibility. They are considered optional integrations unless explicitly identified as part of the current implementation.

API / Service	Use in Quivaultis	Technical Requirements	Feasibility / Current Scope
RAWG API or IGDB API	Optional game metadata enrichment for titles, genres, tags, platforms, ratings, cover images and release information.	API key or OAuth access. Requests should be made from the backend and useful metadata can be cached in MongoDB.	Feasible as a future enhancement. The current prototype can use MongoDB game records without depending on an external game-data API.
SSLCOMMERZ Payment Gateway	Future live checkout/payment confirmation for a Bangladesh-based payment flow.	Merchant sandbox/live credentials plus backend success, fail, cancel and IPN routes. Payment must be verified before granting ownership.	Feasible, but not claimed as implemented in the current prototype. The current store uses an internal demonstration purchase action.
Cloudinary Upload API	Optional hosted storage for avatar, marketplace images and support attachments.	Cloudinary credentials, upload policy and file type/size validation. Signed backend uploads are preferred for sensitive operations.	Feasible as a future storage improvement. It avoids storing large binary data directly in MongoDB.
Internal Recommendation Logic	Mood-based game suggestions using mood values, play preferences, stored game tags/genres and user feedback.	No paid AI API is required. The recommendation logic can score MongoDB game records and store mood/feedback history.	Highly feasible and appropriate for the academic prototype because it keeps the core recommendation feature under project control.

Feasibility Conclusion

The core MERN implementation is technically feasible without paid external services. MongoDB Atlas, REST APIs, JWT-based authentication and internal recommendation logic are sufficient for the main academic demonstration. External services can be integrated later when live payment, hosted media storage or richer third-party game metadata are required.

Module 1

Member	Feature	Problem Identification	Clarification	Specification
Suailim Islam Sakif	Online Game Store, Search, Comparison and Cart System	Customers may have difficulty finding and evaluating suitable games when the catalogue contains many options. A simple list does not help users narrow results, compare alternatives or organize items before purchase.	Customers can browse the store, search by game name, filter by genre, playtime, maximum price and discount, open detailed information, select exactly two games for comparison, and manage selected games through the shopping cart.	The system shall display available games with title, genre, price, discount, rating, playtime/platform information and description. It shall support title search, multiple filters, a game-details view, exactly-two-game comparison, add/remove cart operations, quantity changes and cart clearing. An authenticated customer may complete the internal purchase flow so the purchased games can be associated with the customer's Owned Games Library.
Maisha Thasin	Peer-to-Peer Marketplace System	Players may want to sell unused games or skins and buy items from other players. Without an in-platform marketplace, these exchanges are difficult to organize and users cannot manage listings, offers, inventory, wallet balance and transaction activity in one place.	The marketplace provides Marketplace, Sell, My Listings, Inventory and Transactions views. Customers can search active listings, filter by item type, platform and maximum price, sort results, create listings, send offers, buy eligible listings with wallet balance, delist their own items and manage purchased inventory.	The system shall require authentication for marketplace operations. It shall allow customers to create listings with type, platform, title, valid price, condition, tags and description; display active listings; and support search, filtering and sorting. The system shall prevent users from buying their own listing, verify sufficient wallet balance, deduct the buyer amount, credit the seller, mark the listing as sold, transfer the item to buyer inventory and record Purchase/Sale transactions. It shall also support offer submission, owner-only delisting and marketplace summary information.
Ohidul Hossain	Community Review System	Customers need trustworthy experience-based feedback before choosing a game. If reviews are not connected to real ownership or cannot be managed properly, the review section may contain low-quality or unreliable feedback and users may struggle to identify useful opinions.	Only games in the customer's Owned Games Library are available for review. A customer can give a 1-5 star rating, write a review, edit or delete their own review, mark other reviews as helpful, react to reviews, search by review text/game/user, filter by game, sort community feedback and view review statistics and a highlighted top review.	The system shall require authentication and verify game ownership before creating or updating a review. Review text shall be between 10 and 400 characters and rating shall be between 1 and 5. A customer shall have only one review per game and may edit or delete only their own review. The system shall support helpful toggling, supported reactions, search, game filtering, sorting by top/new/high/low rating, overall review statistics and top-review presentation.
Mutasim Ahsan	Customer Profile Management	Customers need one reliable place to maintain personal information and gaming preferences used throughout the platform. Without structured profile management, account information can become incomplete or inconsistent and profile-completion-based badge eligibility cannot be determined clearly.	Customers can load and update their saved profile, maintain contact/location details, platform and weekly playtime, favorite genres, gamer tag and biography, monitor completion percentage and rank, view badge status, reset unsaved changes and export their current profile information.	The system shall allow an authenticated customer to view and update full name, username, email, phone, country, city, date of birth, platform, weekly playtime, favorite genres, gamer tag and bio. It shall calculate profile completion percentage and assign ranks such as Starter, Bronze, Silver, Gold and Elite. The profile shall display badge status and last-updated information, support resetting unsaved changes, and allow the customer to export profile data in a spreadsheet-compatible file.

Implementation note: The current Store uses an internal purchase flow. A live payment gateway is intentionally treated as a future integration.

Module 2

Member	Feature	Problem Identification	Clarification	Specification
Maisha Thasin	Mood-Based Game Recommendation System	Customers may want to play a game but may not know which title best fits their current mood, available session time, preferred play style or desired difficulty. A normal catalogue can show many games, but it does not adapt recommendations to the user's immediate situation.	Customers can choose a mood archetype or adjust mood values, select session duration, play style and difficulty, and then generate ranked recommendations. Results show a best match together with MoodFit/score information. Customers can respond with Love, Played or Not for Me and can reload previous preference choices.	The system shall accept mood and play-preference input, evaluate available game data and generate ranked game suggestions. It shall display the best match and additional recommendations with relevant match/score information, genre, playtime, rating, tags, discount and price. The system shall retain recent mood/preferences and recommendation feedback, and it shall allow Love, Played and Not for Me actions so later recommendations can reflect stored user taste.
Suailim Islam Sakif	Staff Game Management System	A digital game store needs accurate game data, stock and discount information. Without dedicated maintenance, customers may see outdated or incorrect catalogue information.	Authorized staff can search game records, add new games, edit existing information, update stock/discount values and remove games when required.	The system shall allow authorized Staff/Admin users to create, read, update and delete game records. Editable data shall include title, studio, genre, platform, playtime, base price, discount, rating, difficulty, stock, sold count, image data, description, tags and mood tags. Protected backend routes shall require authentication and appropriate role access.
Ohidul Hossain	Support Ticket and Issue Management System	Customers may face problems with game keys, disks, installation, orders, accounts, wallets or other product-related activities. Without a structured support process, issues may be lost, repeated or difficult for customers and staff to track from submission to resolution.	Customers can create tickets with detailed issue information, optionally attach an image, search/filter/sort their own tickets, open ticket details, send follow-up messages and request reopening. Staff can view submitted tickets, communicate with customers through threaded replies and update the ticket progress.	The system shall allow customers to submit issue type, category, order ID, product name, platform, region, priority, description, contact email and attachment information. Each ticket shall have a trackable ticket code and support threaded messages. Ticket status shall use Pending, Processing and Completed. Staff shall be able to update status and reply; customers shall be able to view their own tickets, reply and request reopening when a completed issue still requires attention.
Mutasim Ahsan	Admin User and Platform Management	The platform requires centralized control over user accounts, role-based access and key operational records. Without an administrative management area, privileged accounts and platform activities cannot be supervised consistently and incorrect or unnecessary records may remain unmanaged.	The Admin interface provides an operational overview and user-management workspace. Admin can view and search users, create Customer, Staff or Admin profiles, delete eligible users and open management sections for games, deals, badge requests and support tickets. The overview also presents useful operational counts.	The system shall require an authenticated Admin role for administrative operations. Admin shall be able to view user records, search by username, email or role, create role-based profiles with the required account fields, and delete users while preventing deletion of the currently logged-in admin account. The interface shall expose operational counts and navigation to game, deal, badge-request and ticket management so core platform records can be supervised from one administrative area.

Module 3

Member	Feature	Problem Identification	Clarification	Specification
Maisha Thasin	Owned Games Library	Customers need a dedicated place to identify games already associated with their account. Without a personal library, purchased games can be difficult to distinguish from titles that are still only available in the store.	After the current internal purchase flow, the customer can open the Owned Games page and retrieve only the games linked to the authenticated account. The library gives the customer a simple personal view of owned titles rather than mixing them with the full store catalogue.	The system shall require authentication before loading owned games and shall retrieve only games associated with the logged-in user. It shall display relevant information for each owned title, including game name/title, genre, rating, playtime, price or owned status and description where available. The library shall be updated when the store purchase flow successfully associates new games with the customer account.
Ohidul Hossain	Wallet and Marketplace Transaction Tracking System	Peer-to-peer trading requires accurate wallet updates and a clear history of marketplace activity. Without balance validation and transaction records, buyers and sellers cannot reliably understand how purchases and sales changed their wallet or confirm what happened to a listed item.	The marketplace shows wallet balance together with inventory and active-listing information. During a purchase, the system checks the buyer's balance, updates both buyer and seller balances, transfers the item into the buyer's inventory and keeps a transaction record that can later be viewed from the Transactions section.	The system shall verify that a listing is active, prevent a seller from buying their own listing and confirm sufficient buyer balance. On success it shall deduct the listing price from the buyer, credit the seller, mark the listing as sold, record the buyer, transfer the purchased item to buyer inventory and create separate Purchase and Sale transaction records. Transaction history shall be available only to the authenticated user and shall contain transaction type, item title, amount, note and creation time.
Suailim Islam Sakif	Badge Verification System	Customers who complete their gamer profile need a controlled way to request verification. Without a badge workflow, the platform cannot distinguish pending, approved and rejected verification states.	Profile completion determines eligibility. At 100% completion, a customer can submit a badge request. Authorized Staff or Admin users can review pending requests and approve or reject them.	The system shall prevent ineligible badge requests, create a Pending request for an eligible customer, allow Staff/Admin users to view requests, approve or reject pending requests and update the customer badge status. The profile shall display badge state such as None, Pending, Approved or Rejected.
Mutasim Ahsan	Deals and Watchlist Management System	Customers may miss useful discounts when reduced-price games are mixed with the full store catalogue. They also need a simple way to remember interesting deals and control their preferred alert threshold without repeatedly searching the store for the same games.	The Deals page shows only games with an active discount and presents useful information such as original price, discounted price, savings, genre, rating and playtime. Customers can search discounted titles, sort by best deal, newest, lowest/highest price or rating, watch or unwatch games, clear the watched list, open the Store and configure price-drop alert preferences.	The system shall load discounted games from the existing store data and allow customers to search and sort the deal list. Authenticated customers shall be able to watch or unwatch a game, retrieve their watched deals and clear the watched list; watched items shall be stored for the logged-in customer. The current implementation shall also allow users to enable or disable deal-alert preferences and select a discount threshold, store those preferences client-side, and create dashboard notifications for supported watchlist and settings actions.

Final Module Distribution

Member	Module 1	Module 2	Module 3
Suailim Islam Sakif	Online Game Store, Search, Comparison and Cart System	Staff Game Management System	Badge Verification System
Maisha Thasin	Peer-to-Peer Marketplace System	Mood-Based Game Recommendation System	Owned Games Library
Ohidul Hossain	Community Review System	Support Ticket and Issue Management System	Wallet and Marketplace Transaction Tracking System
Mutasim Ahsan	Customer Profile Management	Admin User and Platform Management	Deals and Watchlist Management System

References for API Feasibility

- **RAWG API Documentation:** <https://rawg.io/apidocs>
- **RAWG API Reference:** <https://api.rawg.io/docs/>
- **IGDB API Documentation:** <https://api-docs.igdb.com/>
- **Cloudinary Upload API Reference:** https://cloudinary.com/documentation/image_upload_api_reference
- **SSLCOMMERZ Developer Documentation:** <https://developer.sslcommerz.com/doc/v4/>